Sharif University of Technology
Department of Chemical Engineering

Mathematical Modeling and Process Simulation of CO2 Removal from Natural Gas using Hollow Fiber Membranes

Course: Chemical Engineering Unit Operations 2

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July 31, 2026

Abstract

Membrane gas separation, particularly using hollow fiber modules, is an energy-efficient alternative to conventional amine absorption for natural gas sweetening. In this project, a comprehensive mathematical model was developed in Python to simulate multicomponent gas transport through hollow fiber membranes. The stiff system of nonlinear ordinary differential equations (ODEs) was solved using the Orthogonal Collocation method, avoiding the computational inefficiency of traditional trial-and-error shooting methods. To ensure numerical stability, boundary singularities at the closed permeate end were resolved using L'Hôpital's rule, and a Homotopy continuation technique was applied. The model successfully reproduces parametric studies and provides an adaptive solver to optimize module dimensions for achieving pipeline specifications (2.5 mol% CO2).

1 Introduction

Natural gas is a primary global energy source, but raw natural gas often contains significant amounts of CO2 that must be removed to meet grid specifications. Membrane technology offers a robust, small-footprint, and environmentally friendly alternative to chemical absorption. Determining the precise concentration and pressure profiles along the length of hollow fiber membranes is crucial for designing efficient modules.

This project computationally reproduces and enhances the membrane model presented by Chu et al. (2019). The simulation is implemented in Python, utilizing advanced numerical techniques from the `scipy.optimize` library to evaluate the performance of Cellulose Acetate (CA), Polyimide (PI), and Carbon membranes.

2 Mathematical Modeling

The mathematical model simulates co-current and counter-current flow patterns. The core assumptions include uniform axial flows, isothermal conditions, and plug flow behavior in both the shell and bore sides.

The governing differential equations for flow rates are defined as follows:

$$\text{Feed side:} \quad \frac{d(u_{xi})}{dz} = \pm \pi D_o N Q_i (P x_i - p y_i) \tag{1}$$

$$\text{Permeate side:} \quad \frac{d(v_{yi})}{dz} = \pi D_o N Q_i (P x_i - p y_i) \tag{2}$$

Where u_{xi} and v_{yi} are the flow rates of component i, P and p are the pressures on the feed and permeate sides, and Q_i is the permeance.

To reduce mathematical complexity, these equations are non-dimensionalized using reference parameters (e.g., $z^* = z/L$, $P^* = P/P_f$). The dimensionless groups, particularly the pressure drop coefficients (K_{feed} and K_{perm}), are calculated dynamically based on the flow configuration.

3 Numerical Methodology

The multicomponent gas transport equations present a highly stiff boundary value problem (BVP). To achieve a robust solution, the following algorithms were implemented in Python:

3.1 Orthogonal Collocation

Instead of solving the ODEs step-by-step, the Orthogonal Collocation method was used. The domain is mapped onto the roots of Legendre polynomials, and the differential equations are transformed into a system of nonlinear algebraic equations using a Lagrange differentiation matrix (Fornberg algorithm).

3.2 L'Hôpital's Limit for Boundary Singularity

In operations without a sweep gas, the permeate velocity at the closed end ($z = 0$) is zero, which causes a mathematical singularity ($\frac{0}{0}$) when calculating the permeate mole fraction (y_i). To maintain physical and mathematical validity, L'Hôpital's rule was explicitly programmed at the boundary to evaluate the limiting flux ratios, completely eliminating artificial concentration gradients.

3.3 Homotopy Continuation

Newton-Raphson solvers typically fail to converge for stiff membrane equations. A permeance-ramping Homotopy method was developed, starting with an impermeable membrane (linear solution) and gradually scaling the permeance ($Q\_scale \to 1.0$). The `scipy.optimize.root` hybrid and Levenberg-Marquardt solvers were used.

4 Results and Discussion

4.1 Module Design Parametric Studies

The developed Python simulator, *Mollocator*, was utilized to systematically investigate the effect of geometric parameters on module efficiency.

As shown in Figure 2, the bore-side pressure drop is highly dependent on the hollow fiber inner diameter. Significant pressure drops are observed for fibers with $D_i < 200\mu m$.

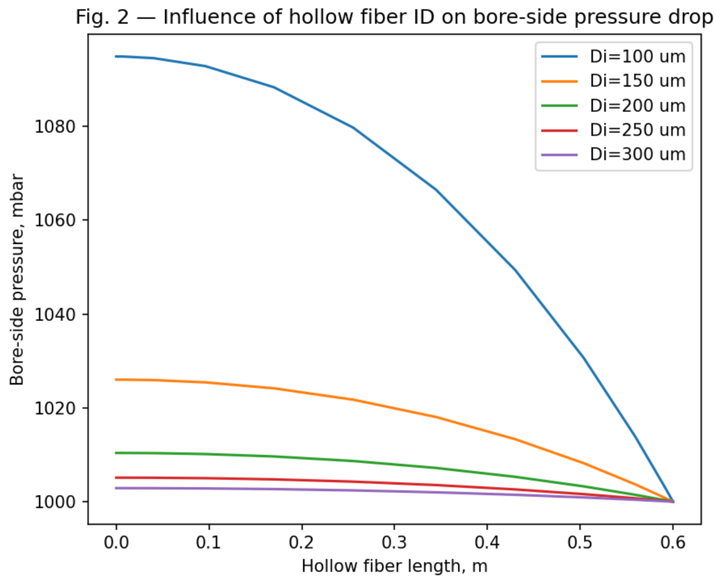

Figure 1: Influence of hollow fiber inner diameter on bore-side pressure drop.

Figures 3 and 4 demonstrate the effects of fiber length and packing density, respectively. A packing density that is too high causes an extreme increase in shell-side pressure drop.

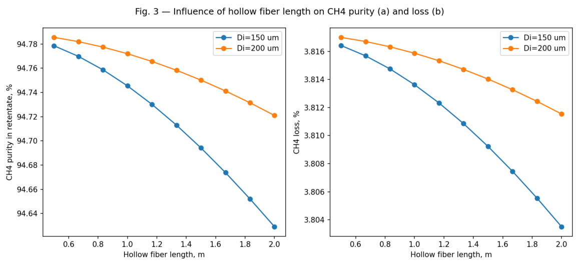

Figure 2: CH4 purity and loss vs. fiber length.

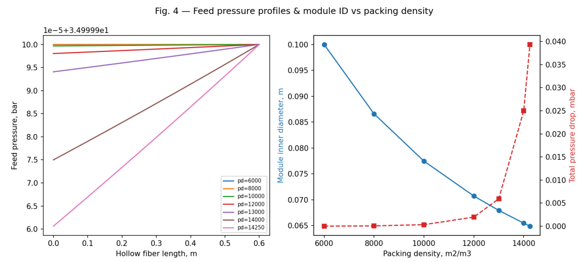

Figure 3: Module ID and pressure drop vs. packing density.

4.2 Natural Gas Sweetening Process Study

The model was extended to an adaptive solver that uses a bisection search (`scipy.optimize.brentq`) to automatically size the module (find the optimal number of fibers, N) to achieve a target retentate CO_2 concentration of 2.5 mol%.

Figure 5 visualizes the dimensionless concentration and absolute pressure profiles along the length of the fibers for three different membrane materials (CA, PI, Carbon).

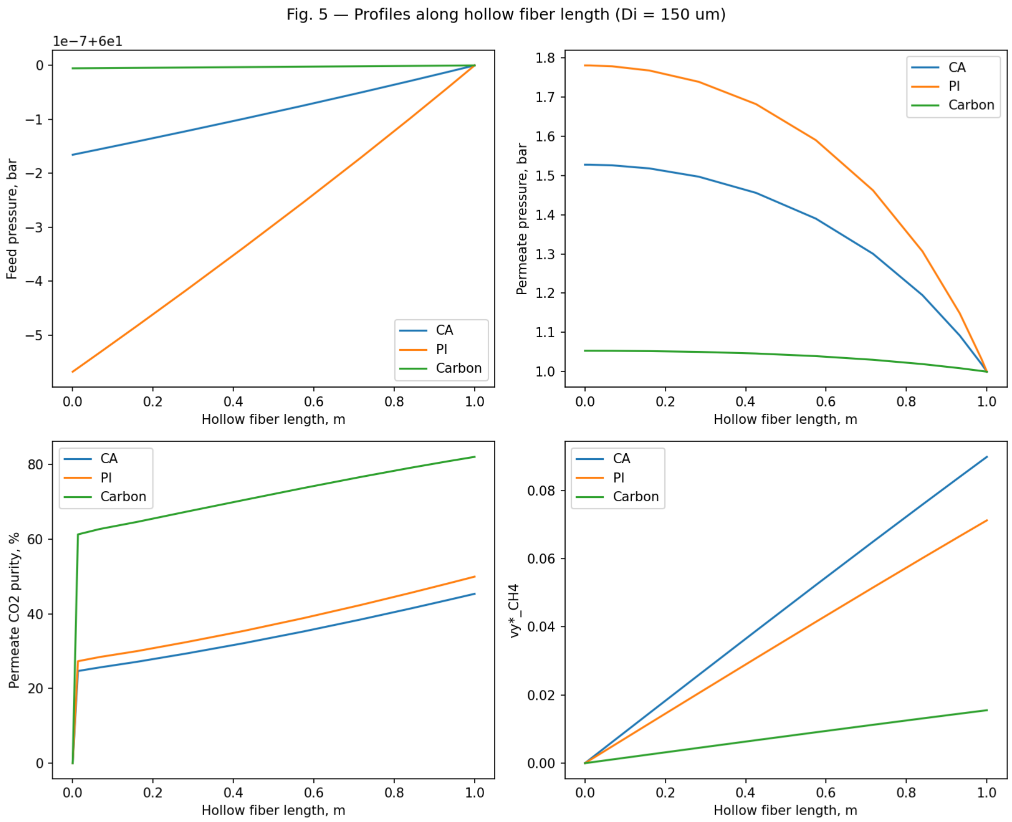

Figure 4: Concentration and pressure profiles along the hollow fiber length.

Further parametric sweeps (Figures 6 and 7) reveal that while Carbon membranes require a larger specific membrane area due to lower permeance, they significantly minimize hydrocarbon (HC) loss compared to polymeric membranes.

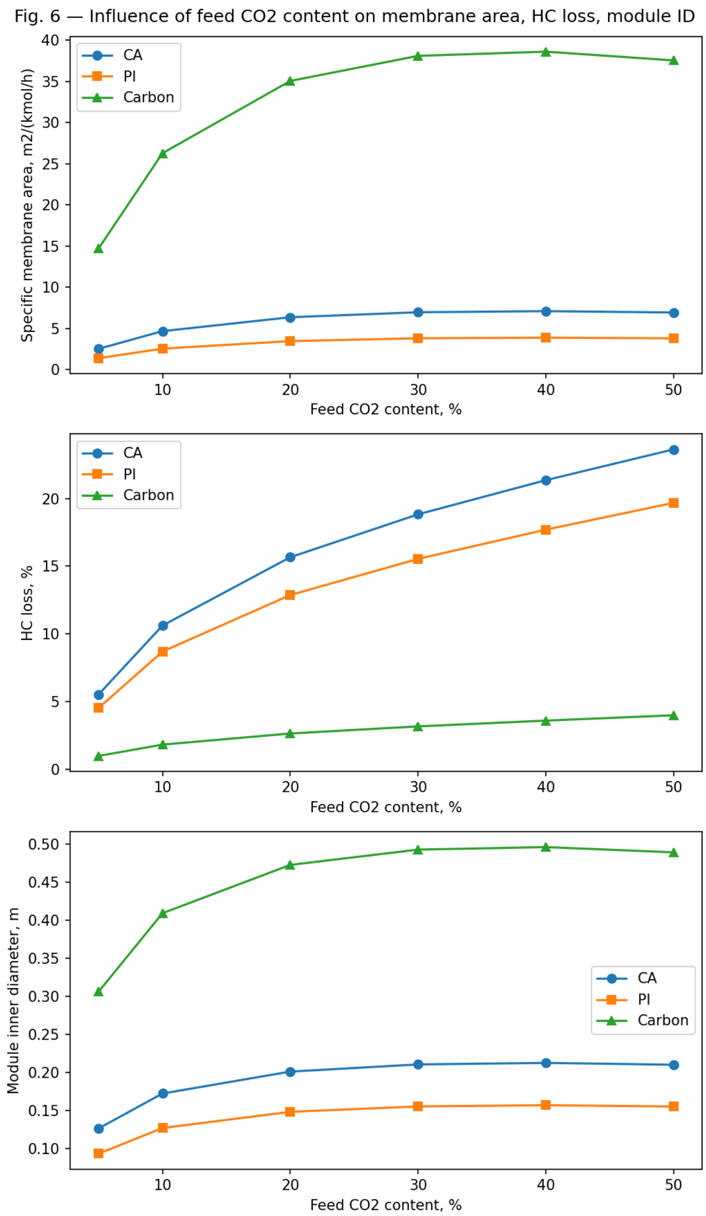

Figure 5: Influence of feed CO2 content on specific membrane area and HC loss.

5 Conclusion

A highly accurate, object-oriented Python simulator was successfully developed to model multi-component gas separation in hollow fiber membranes. By implementing Orthogonal Collocation alongside rigorous boundary condition treatments, the numerical instability associated with stiff transport equations was mitigated. The model serves as a robust tool for process simulation, module design, and techno-economic evaluation of gas sweetening plants.

References

[1] Chu, Y., Lindbråthen, A., Lei, L., He, X., & Hillestad, M. (2019). *Mathematical modeling and process parametric study of CO2 removal from natural gas by hollow fiber membranes.* Chemical Engineering Research and Design, 148, 45-55.